

Obaidullah Nasir

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Education

MS Engineering Management

CASE Institute of Technology, Islamabad

2022 - 2024

BE Electrical Engineering

National University of Sciences and Technology (NUST) Islamabad, Pakistan

CGPA: 3.52/4.00 (**High Achiever Award Recipient**)

2016 - 2020

Experience

VisionRD – AI Engineer

June 2025 - Current

- Developed and deployed vFuseAct, a **dual-head MemoryTCN model** with temporal segmentation for real-time, framewise action segmentation on Raspberry Pi Zero 2W at 30 FPS. Used in HAVAL's automotive plant to validate assembly steps live, reducing **human errors by 33%** and increasing technician **onboarding time by 70%**.
- Developed a Forgery Detection System using OCR+LLM, deployed via cron job at a **Toyota plant** (Karachi, Pakistan), featuring fallback logic to detect tampering and parse noisy scanned documents with stamps, barcodes, and handwritten or missing text, achieving **98% accuracy** and a **70% reduction in manual verification**.
- Engineered a production-grade **RAG-based marketing tool** for Sazgar Motors (Pakistan's major automotive plant), leveraging Selenium for dynamic data ingestion and Groq API for high-speed LLM inference, enabling real-time conversational analysis of consumer sentiment, reducing **manual analysis by 80%**.
- Implemented the GriTS evaluation framework in partnership with **Landing AI** to benchmark table structure recognition across multiple document layout models, optimizing structure prediction metrics for downstream parsing.
- Built a PCB Defect Detection System using custom computer vision architectures to identify faulty components and soldering faults in Rapidev's (Pakistan's major PCB manufacturing Company) production line, inspecting 10–15 million PCBs per year in real time.
- Lead continuous **R&D initiatives** to evaluate emerging architectures in Computer Vision and NLP, conducting rapid feasibility studies and prototyping to integrate cutting-edge research into production workflows.

Ministry of Defense - Design Engineer

Nov 2020 – Jan 2025

- Design of military-grade Laser Actuated Target System for realistic military training, using ARM-Cortex-M3, LRF modules, and LiDAR-Lite v3HP optical sensor, resulting in a **\$76,000 annual decrease** in training ammunition expenditure.
- Development of a cost-effective **training simulator** using Bohemia Interactive's Real Virtuality 4 physics engine, optimizing realistic simulations for enhanced training effectiveness, and reducing **annual training expenditure by 40%**.
- Developed an **offline RAG system** for military document intelligence, using LangChain, LLaMA 3, and ChromaDB to enable secure, air-gapped conversational search over SOPs, technical manuals, and classified PDFs, optimized for high accuracy and low-latency response in constrained, disconnected environments.
- Design of cost-effective EV Cart, using Solar technology, integrated with IoT and Battery Management System.
- Implemented sensor fusion (Machine Vision and LiDARs) for real-time inventory tracking and automated logistics planning, optimizing spare parts availability for the military fleet (3000+ vehicles).

Technical Project Manager

Oct 2021 – Aug 2024

- Pivotal role in forecasting high-spend projects, budgeting, design supervision, cost, and data analysis.
- Responsible for administering 250 technicians along with technical projects and repair line management of diverse vehicles and machinery, boosting **workshop productivity by 30%**.
- Led installation of solar-powered reverse osmosis plants in flood-affected remote areas, providing essential clean water access to victims

Certifications

- Machine Learning, Deep Learning, and RAG Specialization** – DeepLearning.AI
- AWS Cloud Technical Essentials - AWS

Technical Skills

- Programming Languages: Python, HTML, CSS, SQL
- Software & Tools: PyCharm, VSCode, MS Excel, LabView
- DevOps & Cloud Technologies: Docker, Version Control (Git), CI/CD Workflows, AWS
- Databases & ORM: SQL, SQLAlchemy, ChromaDB
- Hardware Expertise: Arduino, PIC Microcontrollers, LiDAR Sensors

Personal Interests and Skills

- Leadership and teamwork
- Commitment, problem solving and enthusiasm
- Self-management (including timekeeping)
- Sports, fitness and travelling